Service Tool Instruction

Service Tool Instruction Number: 3377947	Revision Level:	Released Date: 18-jun-2003

Description

Crankshaft Wear Sleeve Installer

Purpose

This document provides information for using the crankshaft wear sleeve installer, Part Number 3165112. The front crankshaft wear sleeve/seal kit, Part Number 3925343, is used to salvage the crankshaft with grooves worn in the oil seal area on C8.3, C8.3G, ISC, QSC8.3, ISL, and QSL9 engines. This tool installs the front crankshaft wear sleeve to the proper depth.

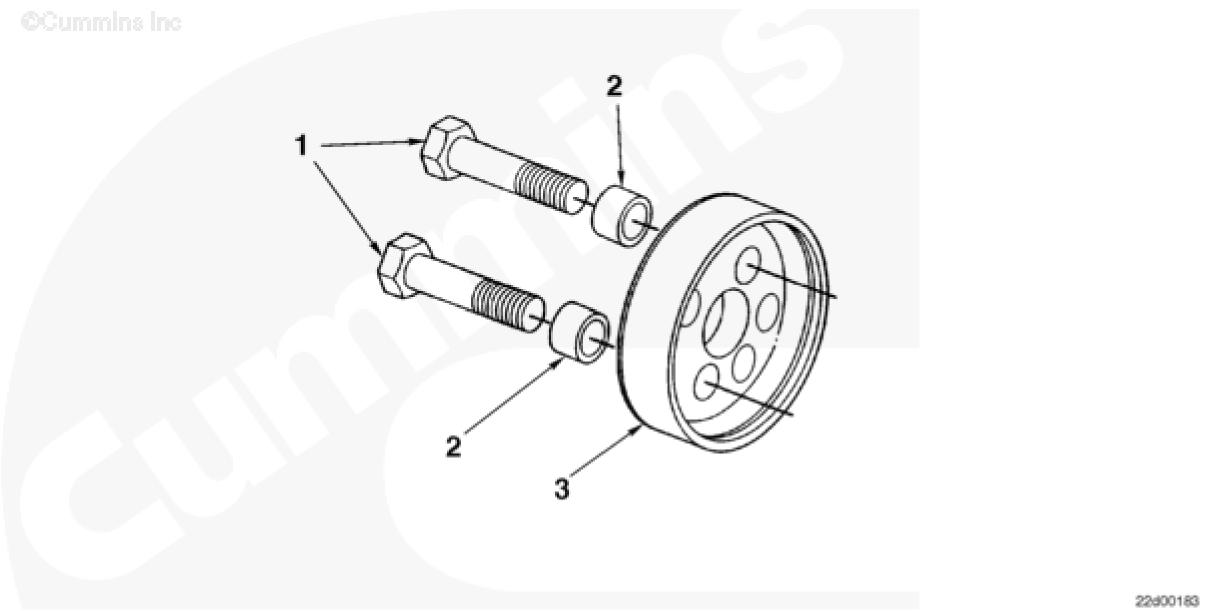


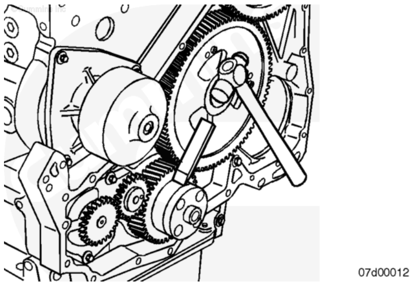
Table 1, Crankshaft Wear Sleeve Installer, Part Number 3165112			
Item Number	**Part Number**	**Description**	**Quantity**
1	3164598	Capscrew	3
2	3164599	Spacer, 5/8-inch length	3
3	3165111	Installer plate	1

Note : The previous installer, Part Number 3824501, installed the 0.984-inch wide wear sleeve on C8.3, C8.3G, ISC, and QSC8.3 engines. Part Number 3165112

supersedes the previous installer and is used on ISL and QSL9 as well as C8.3, C8.3G, and ISC engines.

⚠ CAUTION ⚠

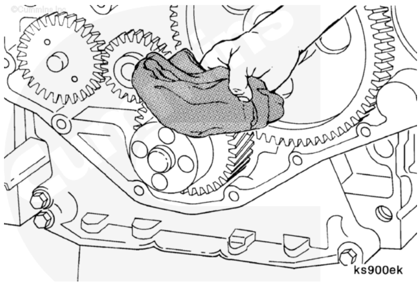
Do not nick or gouge the crankshaft with the chisel. If the crankshaft is damaged, it must be replaced.



Remove the seal carrier and/or front cover, refer to the appropriate Troubleshooting and Repair Manual.

If a wear sleeve is present, use a hammer and a bull chisel **only** as wide as the wear sleeve. Make one or two blows with a hammer to make chisel marks across the wear sleeve. Repeat this procedure at two or three locations around the circumference of the sleeve. This will expand the wear sleeve, allowing the sleeve to be removed.

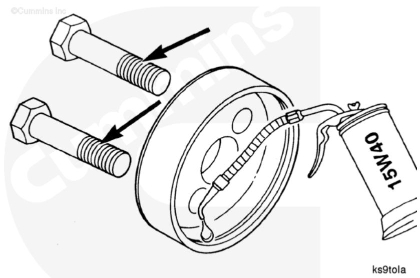
For proper sealing of the crankshaft oil seal, the crankshaft **must** be free of major damage.



Use a crocus cloth to remove any rust or other deposits from the crankshaft flange area.

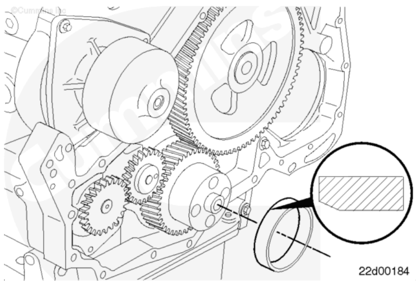
Use a clean cloth to clean the crankshaft flange area.

Note : Three capscrews are required for ISL and QSL9 engines.



Apply a thin coat of clean 15W-40 engine oil on the inside diameter of the installer plate and capscrew threads.

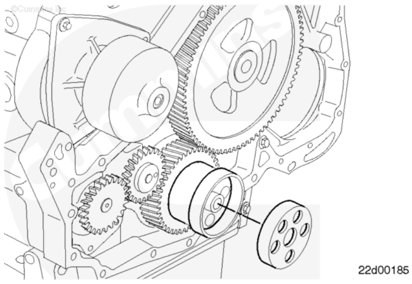
Gasket sealant between the wear sleeve and crankshaft is **not** recommended.



Apply a thin coat of clean 15W-40 engine oil on the crankshaft flange.

Position the chamfered end of the wear sleeve onto the end of the crankshaft.

Position the counterbore end of the installer plate onto the wear sleeve.

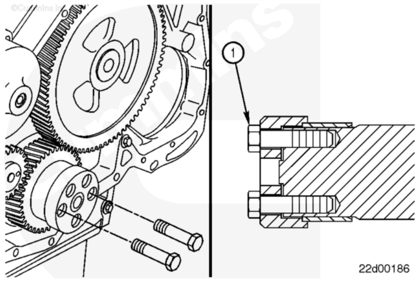


Note : Three capscrews are required for ISL and QSL9 engines.

Install the capscrews (1), without the spacers, through the installer plate and into the crankshaft holes.

Align the wear sleeve and installer plate perpendicular with the crankshaft.

Finger tighten the capscrews.



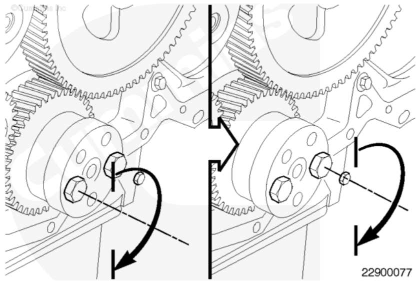
To prevent wear sleeve damage from binding and irregular stretch, do **not** exceed 1/2 revolution on each capscrew.

Alternately tighten the capscrews until the wear sleeve is installed approximately 16 mm [0.63 in].

Remove the capscrews and add the spacers.

Continue to alternately tighten the capscrews until the installer plate contacts the end of the crankshaft. Do **not** exceed the recommended torque value.

Torque Value: 20 n•m [15 ft-lb]



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